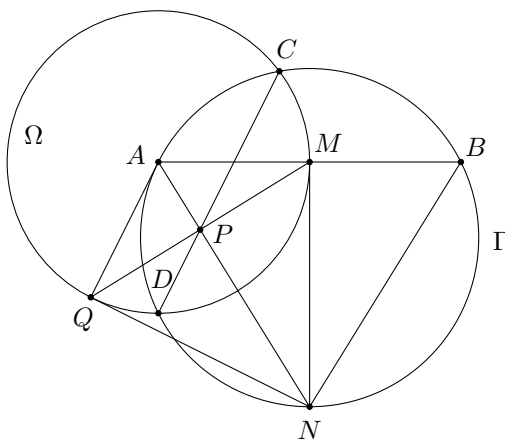


Problem 53 (Singapore MO Open 2010 Problem 1). Let AB be a chord of a circle Γ . Let M be the midpoint of AB . The circle Ω with centre A and passing through M meets Γ at C and D . Let N be the midpoint of the major arc AB in Γ . The segment AN meets CD at P . The line MP meets Ω again at Q . Prove that CD is perpendicular to NQ .



Solution. We have $PA \times PN = PC \times PD = PM \times PQ$. This implies A, Q, N, M are concyclic. Note that $\angle AMN = 90^\circ$ since MN is the perpendicular bisector of AB . Thus, $\angle AQN = 90^\circ$, and hence NQ and NM are tangents to Ω . It follows that $\angle QAN = \angle MAN = \angle ABN$, showing that AQ is tangent to Γ . As A is the midpoint of arc CD , we have $AQ \parallel CD$. Together with $AQ \perp NQ$, we obtain $CD \perp NQ$. $\square$

Problem 54 (Polish MO 2018 Second Round Problem 6). Let n be a positive integer. Suppose $\{x_k\}$ is a sequence of integers such that $0 \leq x_j \leq n$ for all j . Define another sequence $\{y_k\}$ by

$$y_k = \sqrt[k]{x_1^k + x_2^k + \cdots + x_n^k}.$$

If there are infinitely many terms in $\{y_k\}$ which are integers, prove that all the terms in $\{y_k\}$ are integers.

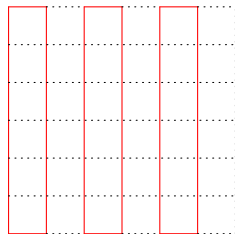
Solution. Let $M = x_a$ be one of the largest terms in the sequence $\{x_k\}$. For any sufficiently large $k \geq a$, we have $y_k \geq \sqrt[k]{x_a^k} = M$ and

$$y_k \leq \sqrt[k]{M^k + M^k + \cdots + M^k} = \sqrt[k]{k}M < M + 1$$

since $\lim_{k \rightarrow \infty} \sqrt[k]{k} = 1$. As there are infinitely many k such that y_k is an integer, there are infinitely many such k with $M \leq y_k < M + 1$. This means $y_k = M$. This holds when $x_j = 0$ for all $j = 1, 2, \dots, k$ with $j \neq a$. As there are infinitely many such k , this means $x_j = 0$ for all j except possibly $j = a$. In that case it is obvious that $y_k = 0$ whenever $k < a$, and $y_k = M$ whenever $k \geq a$, all of which are integers. $\square$

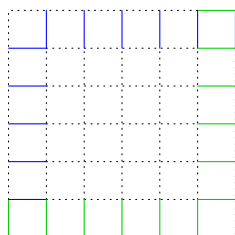
Problem 55 (Singapore MO Open 2008 Problem 5). There is a 2022×2022 grid where the grid lines are white in colour. We draw n rectangles in red colour using some of the grid lines as sides. It turns out that all the grid lines of the grid become red. Find the smallest possible value of n .

Solution. The answer is 2023.



We provide the construction as follows. Firstly, we draw 1011 rectangles of size 2022×1 from the left to the right, so that two consecutive rectangles are separated by 1 unit. Then all vertical grid lines except the last become red. Secondly, we draw 1011 rectangles of size 1×2022 from the top to the bottom in the same way. Then all horizontal grid lines except the last become red. Lastly, we draw one more rectangle of size 2022×2022 to cover the remaining two grid lines. This gives a construction with $1011 + 1011 + 1 = 2023$ rectangles.

Next, we prove that the bound is optimal. Consider the 2022 unit segments in the first row except the leftmost one, and the 2022 unit segments in the first column except the topmost one. We call them blue segments. It is not hard to see that each rectangle only contains at most two blue segments, and so at least $\frac{2022 + 2022}{2} = 2022$ rectangles should be drawn. It suffices to consider the case when exactly 2022 rectangles are drawn. In that case, each rectangle contains exactly two blue segments and no two rectangles share the same blue segment. Similarly, we call the 2022 unit segments in the last row except the rightmost one, and the 2022 unit segments in the last column except the bottommost one green segments. Then each rectangle contains exactly two green segments and no two rectangles share the same green segment.



Note that there is a rectangle which contains the leftmost segment in the first row. In other words, there is a rectangle R_1 with the top left vertex of the grid as one of the vertices. In order that R_1 contains two blue segments and two green segments, it must be of size $2022 \times k$ or $k \times 2022$ for some k . WLOG assume R_1 has size $2022 \times k$.

If $k < 2022$, then R_1 does not contain the rightmost segment in the last row. Applying the same argument, we see that there is another rectangle R_2 of size $2022 \times \ell$ with the bottom right vertex of the grid as one of the vertices (it cannot have size $\ell \times 2022$ since otherwise R_2 shares the same blue segment as R_1).

Ignoring the blue segments contained by R_1 and R_2 , there are 2021 blue horizontal segments left. Thus, there is another rectangle R_3 containing a blue horizontal segment and a blue vertical segment different from the rightmost one. But then R_3 cannot contain any green segments, which is a contradiction.

Similarly, if $k = 2022$, by ignoring the blue segments contained by R_1 , there are 2021 blue horizontal segments left. Thus, there is another rectangle R_4 containing a blue horizontal segment and a blue vertical segment different from the rightmost one. But then R_4 cannot contain any green segments, which is a contradiction.

Thus, we need to draw at least 2023 rectangles as desired. □

Problem 56 (China TST 2009 Quiz 6 Problem 6). Let $\{a_n\}$ be a sequence of positive integers such that $\gcd(a_m, a_n) = a_{\gcd(m, n)}$ for any positive integers m and n . Prove that $\prod_{d|n} a_d^{\mu(\frac{n}{d})}$ is an integer for every positive integer n , where the product runs through all positive divisors d of n . (Here, $\mu(m)$ is the Möbius function.)

Solution. Let $f(n) = \prod_{d|n} a_d^{\mu(\frac{n}{d})}$. Then $a_n = \prod_{d|n} f(d)$ by the product version of the Möbius inversion formula.

Firstly, $f(1) = a_1$ is an integer. Consider any other $f(k)$ where the prime factorization of k is $p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$. Let $q_j = \frac{k}{p_j}$ for $j = 1, 2, \dots, s$. Since $q_j \mid k$, we have $(a_{q_j}, a_k) = a_{q_j}$, and so $a_{q_j} \mid a_k$. Therefore, $L = \text{lcm}(a_{q_1}, a_{q_2}, \dots, a_{q_s})$ divides a_k . Note that

$$\begin{aligned} L &= \left(\prod_{i=1}^s a_{q_i} \right) \left(\prod_{1 \leq i < j \leq s} (a_{q_i}, a_{q_j}) \right)^{-1} \left(\prod_{1 \leq i < j < \ell \leq s} (a_{q_i}, a_{q_j}, a_{q_\ell}) \right) \cdots (a_{q_1}, a_{q_2}, \dots, a_{q_s})^{(-1)^{s+1}} \\ &= \left(\prod_{i=1}^s a_{q_i} \right) \left(\prod_{1 \leq i < j \leq s} a_{(q_i, q_j)} \right)^{-1} \cdots a_{(q_1, q_2, \dots, q_s)}^{(-1)^{s+1}} \\ &= \left(\prod_{i=1}^s \prod_{d \mid q_i} f(d) \right) \left(\prod_{1 \leq i < j \leq s} \prod_{d \mid (q_i, q_j)} f(d) \right)^{-1} \cdots \left(\prod_{d \mid (q_1, q_2, \dots, q_s)} f(d) \right)^{(-1)^{s+1}}. \end{aligned}$$

Note that for any $f(d)$ in the above product, d must be a proper divisor of k . Consider each such divisor $d = p_1^{\beta_1} p_2^{\beta_2} \cdots p_s^{\beta_s}$ where exactly $t > 0$ of the exponents β_i are less than α_i . The exponent of $f(d)$ in the above product is

$$\binom{t}{1} - \binom{t}{2} + \binom{t}{3} - \cdots + (-1)^{t+1} \binom{t}{t} = 1 - (1 - 1)^t = 1.$$

Therefore, we have $L = \prod_{\substack{d \mid k \\ d \neq k}} f(d) = \frac{a_k}{f(k)}$. Given that L divides a_k , it means $f(k)$ is an integer. □